

Let's start w/ Q2:

Consider $1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$ 11-space addition.

2) Determine CG coefficients $\langle \frac{3}{2}, m_{3/2} | m_1, m_{\frac{1}{2}} \rangle$ $\langle \frac{1}{2}, m_{\frac{1}{2}} | m_1, m_{\frac{1}{2}} \rangle$.

$$|j m\rangle = \sum c |j_1 m_1, j_2 m_2\rangle$$

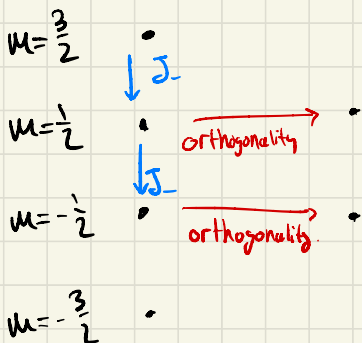
$$j_{\max} = j_1 + j_2 = 1 + \frac{1}{2} = \frac{3}{2}$$

$$j_{\min} = j_1 + j_2 - 1 = j_1 - j_2 = 1 - \frac{1}{2} = \frac{1}{2}.$$

$$j = \frac{3}{2}$$

$$j = \frac{1}{2}$$

dimensionality = 6 ✓



$|\frac{3}{2}, \frac{3}{2}\rangle$ state has $m_1 + m_2 = 1 + \frac{1}{2} = \frac{3}{2}$ which is max

$$\text{Hence } |\frac{3}{2}, \frac{3}{2}\rangle = |1, 1; \frac{1}{2}, \frac{1}{2}\rangle \quad \text{CB}=1$$

$$\text{Similarly } |\frac{3}{2}, -\frac{3}{2}\rangle = |-1, -1; \frac{1}{2}, -\frac{1}{2}\rangle \quad \text{CB}=1$$

$$J_- |\frac{3}{2}, \frac{3}{2}\rangle = \hbar \sqrt{\frac{3}{2}(\frac{3}{2}+1) - \frac{3}{2}(\frac{3}{2}-1)} |\frac{3}{2}, \frac{1}{2}\rangle$$

$$= \sqrt{3} \hbar |\frac{3}{2}, \frac{1}{2}\rangle.$$

$$\begin{aligned}
 &= (J_- |1, 1\rangle) \otimes |\frac{1}{2}, \frac{1}{2}\rangle + |1, 1\rangle \otimes (J_- |\frac{1}{2}, \frac{1}{2}\rangle) \\
 &= \hbar \sqrt{1(1+1) - 1(1-1)} |0, \frac{1}{2}, \frac{1}{2}\rangle + \hbar \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} |1, \frac{1}{2}, -\frac{1}{2}\rangle \\
 &= \sqrt{2} \hbar |0, \frac{1}{2}, \frac{1}{2}\rangle + \hbar |1, \frac{1}{2}, -\frac{1}{2}\rangle
 \end{aligned}$$

Thus, equating $|\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{2}{3}} |0, \frac{1}{2}, \frac{1}{2}\rangle + \frac{1}{\sqrt{3}} |1, \frac{1}{2}, -\frac{1}{2}\rangle$

To determine $|\frac{1}{2}, \frac{1}{2}\rangle$, guess $|\frac{1}{2}, \frac{1}{2}\rangle = a |0, \frac{1}{2}, \frac{1}{2}\rangle + b |1, \frac{1}{2}, -\frac{1}{2}\rangle$

Normalization implies $a^2 + b^2 = 1$, and we require orthogonality w/ the $|\frac{3}{2}, \frac{1}{2}\rangle$ state. $\langle \frac{1}{2}, \frac{1}{2} | \frac{3}{2}, \frac{1}{2} \rangle = 0$ implying $\sqrt{\frac{2}{3}}a + \frac{1}{\sqrt{3}}b = 0$.
 $a^2 + 2a^2 = 1 \Rightarrow a = \frac{1}{\sqrt{3}}$ and $b = -\frac{\sqrt{2}}{\sqrt{3}}$ $\sqrt{2}a = -b$

Thus $|\frac{1}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{3}} |0, \frac{1}{2}, \frac{1}{2}\rangle - \frac{\sqrt{2}}{\sqrt{3}} |1, \frac{1}{2}, -\frac{1}{2}\rangle$.

Determine $|\frac{3}{2}, -\frac{1}{2}\rangle = J_- |\frac{3}{2}, \frac{1}{2}\rangle = \hbar \sqrt{\frac{3}{2}(\frac{3}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} |\frac{3}{2}, -\frac{1}{2}\rangle$
 $= 2\hbar |\frac{3}{2}, -\frac{1}{2}\rangle$.

$$= \sqrt{\frac{2}{3}} \left[(J_- |1, 0\rangle) \otimes |\frac{1}{2}, \frac{1}{2}\rangle + |1, 0\rangle \otimes (J_- |\frac{1}{2}, \frac{1}{2}\rangle) \right] + \frac{1}{\sqrt{3}} \left[(J_- |1, 1\rangle) \otimes |\frac{1}{2}, -\frac{1}{2}\rangle + |1, 1\rangle \otimes (J_- |\frac{1}{2}, -\frac{1}{2}\rangle) \right]$$

$$= \sqrt{\frac{2}{3}} \left[\hbar \sqrt{1(1+1)} |1, -1, \frac{1}{2}, \frac{1}{2}\rangle + \hbar \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} |1, 0, \frac{1}{2}, -\frac{1}{2}\rangle \right]$$

$\frac{3}{4} + \frac{1}{4} = 1$

$$+ \frac{1}{\sqrt{3}} \left[\hbar \sqrt{1(1+1) - 1(1-1)} |1, 0, \frac{1}{2}, -\frac{1}{2}\rangle + \hbar \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} |1, 1, \frac{1}{2}, -\frac{1}{2}\rangle \right]$$

$\frac{3}{4} + \frac{1}{2} \cdot \frac{3}{2} = \frac{9}{4} = \frac{16}{4}$

1/3 cancel

$$2|\frac{3}{2}, -\frac{1}{2}\rangle = \sqrt{\frac{2}{3}} \left[\sqrt{2} |1, -1; \frac{1}{2}, \frac{1}{2}\rangle + |1, 0; \frac{1}{2}, -\frac{1}{2}\rangle \right] \\ + \frac{1}{\sqrt{3}} \left[\sqrt{2} |1, 0; \frac{1}{2}, -\frac{1}{2}\rangle \right]$$

Hence

$$|\frac{3}{2}, -\frac{1}{2}\rangle = \frac{1}{\sqrt{3}} |1, -1; \frac{1}{2}, \frac{1}{2}\rangle + \sqrt{\frac{2}{3}} |1, 0; \frac{1}{2}, -\frac{1}{2}\rangle$$

By orthogonality then,

$$|\frac{1}{2}, -\frac{1}{2}\rangle = a |1, -1; \frac{1}{2}, \frac{1}{2}\rangle + b |1, 0; \frac{1}{2}, -\frac{1}{2}\rangle$$

$$\text{implies } a^2 + b^2 = 1 \quad \text{and} \quad \frac{a}{\sqrt{3}} + \sqrt{\frac{2}{3}} b = 0, \quad a + \sqrt{2} b = 0$$

$$\text{so } 2b^2 + b^2 = 1 \quad \text{implies } b = \frac{1}{\sqrt{3}} \quad \text{and} \quad a = -\sqrt{\frac{2}{3}}$$

$$|\frac{1}{2}, -\frac{1}{2}\rangle = -\sqrt{\frac{2}{3}} |1, -1; \frac{1}{2}, \frac{1}{2}\rangle + \frac{1}{\sqrt{3}} |1, 0; \frac{1}{2}, -\frac{1}{2}\rangle$$

b) The complete CB Matrix:

$$\begin{pmatrix} |\frac{3}{2}, \frac{3}{2}\rangle \\ |\frac{3}{2}, \frac{1}{2}\rangle \\ |\frac{1}{2}, \frac{1}{2}\rangle \\ |\frac{1}{2}, -\frac{1}{2}\rangle \\ |\frac{3}{2}, -\frac{1}{2}\rangle \\ |\frac{3}{2}, -\frac{3}{2}\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{\frac{1}{3}} & \sqrt{\frac{2}{3}} & 0 & 0 & 0 \\ 0 & -\sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{3}} & -\sqrt{\frac{2}{3}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} |1, \frac{1}{2}\rangle \\ |1, -\frac{1}{2}\rangle \\ |0, \frac{1}{2}\rangle \\ |0, -\frac{1}{2}\rangle \\ |-1, \frac{1}{2}\rangle \\ |-1, -\frac{1}{2}\rangle \end{pmatrix}$$

Check its orthogonality:

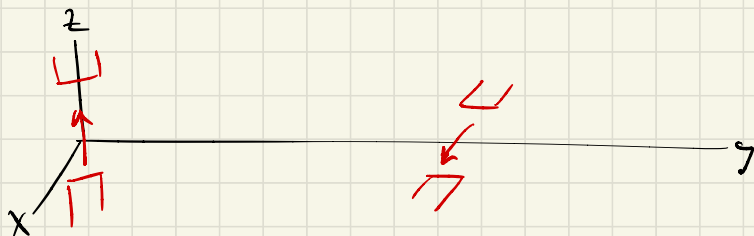
$$O^T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{\frac{1}{3}} & -\sqrt{\frac{2}{3}} & 0 & 0 & 0 \\ 0 & \sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{3}} & -\sqrt{\frac{2}{3}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Hence $O^T O$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{\frac{1}{3}} & -\sqrt{\frac{2}{3}} & 0 & 0 & 0 \\ 0 & \sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{3}} & -\sqrt{\frac{2}{3}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{\frac{1}{3}} & \sqrt{\frac{2}{3}} & 0 & 0 & 0 \\ 0 & -\sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{3}} & -\sqrt{\frac{2}{3}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & & & & & \\ \frac{1}{3} + \frac{2}{3} & & & & & \\ & \frac{2}{3} + \frac{1}{3} & & & & \\ & & \frac{2}{3} + \frac{1}{3} & & & \\ & & & \frac{2}{3} + \frac{1}{3} & & \\ & & & & & 1 \end{pmatrix} = \mathbb{I}_{6 \times 6}.$$

Q6) Stern-Gerlach Apparatus:



Both lower spins blocked.

Rotation

$$\begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi/2} & -\sin \frac{\theta}{2} e^{-i\phi/2} \\ \sin \frac{\theta}{2} e^{i\phi/2} & \cos \frac{\theta}{2} e^{i\phi/2} \end{pmatrix}$$

Incident particles in A_2 is $|+\rangle_z$ ($\theta=0, \phi=0$)

When $\theta = \frac{\pi}{2}$, $\phi=0$, we rotate the state $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ into

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{Hence } |+\rangle_z = \frac{1}{\sqrt{2}}|+\rangle_x + \frac{1}{\sqrt{2}}|-\rangle_x$$

$\uparrow$
x-basis transformation

Measuring $|+\rangle_x$ yields $|_x \langle + | + \rangle_z|^2 = \frac{1}{2}$ (50%)

b) Third filter, again in $+z$.

Taking $|+\rangle_x$ incident and rotate $\theta = -\frac{\pi}{2}$,
 $\downarrow$ z basis transformation

$$\frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow |+\rangle_x = \left(\frac{1}{2}\right)|+\rangle_z - \left(\frac{1}{2}\right)|-\rangle_z$$

$$\text{so } |\langle + | \gamma_x \rangle|^2 = \frac{1}{2},$$

$P(AB) = P(A) \cdot P(B)$ implies 25% of particles leaving first will exit 3rd.

c) If middle filter transmits both spins up/down,

then we rotate $\frac{\pi}{2}$ basis

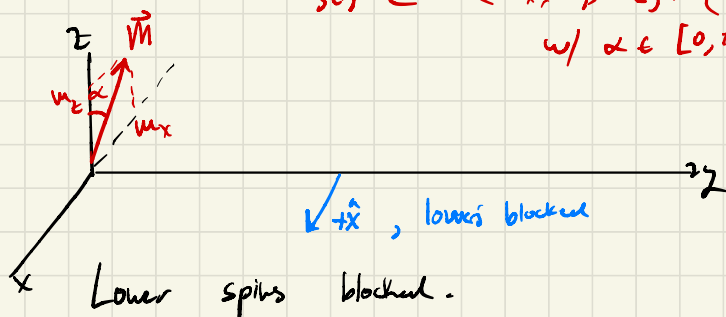
$$\begin{aligned} \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} (1 + \cancel{1}) + \frac{1}{\sqrt{2}} (1 + \cancel{-1}) \right] \\ &= \frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{2}} |+\rangle_z \\ &= |+\rangle_z \end{aligned}$$

nothing changes!

456 Last Years Quiz

Stern-Gerlach:

$$S_y @ (u_x, u_y, u_z) = (-\sin\alpha, 0, \cos\alpha) \\ w/ \alpha \in [0, \pi]$$



Let $\begin{pmatrix} 1 \\ 0 \end{pmatrix}_m$ be the $|+\rangle_m$ state in the $\vec{u}$ basis.

We rotate the state $|+\rangle_m$ into the $\hat{x}$ basis:

$$\varphi = 0, \quad \theta = -\alpha$$

$$\begin{pmatrix} \cos(-\frac{\alpha}{2}) & -\sin(-\frac{\alpha}{2}) \\ \sin(-\frac{\alpha}{2}) & \cos(-\frac{\alpha}{2}) \end{pmatrix} = \begin{pmatrix} \cos\frac{\alpha}{2} & \sin\frac{\alpha}{2} \\ -\sin\frac{\alpha}{2} & \cos\frac{\alpha}{2} \end{pmatrix} \text{ in } \hat{z} \text{ basis,}$$

From which, in the $\hat{x}$ basis, we add $\frac{\pi}{2}$

$$\begin{pmatrix} \cos(\frac{\alpha}{2} + \frac{\pi}{2}) & \sin(\frac{\alpha}{2} + \frac{\pi}{2}) \\ -\sin(\frac{\alpha}{2} + \frac{\pi}{2}) & \cos(\frac{\alpha}{2} + \frac{\pi}{2}) \end{pmatrix}$$

which makes the state in $\hat{x}$

$$|+\rangle_m = \cos\left(\frac{\alpha}{2} + \frac{\pi}{2}\right) |+\rangle_x - \sin\left(\frac{\alpha}{2} + \frac{\pi}{2}\right) |-\rangle_x$$

Projecting along $|+\rangle_x$ yields

$$\langle + | + \rangle_x^2 = \cos^2\left(\frac{\alpha}{2} + \frac{\pi}{2}\right)$$

$$\cos(a+b) = \cos a \cos b - \sin a \sin b$$

$$\cos^2\left(\frac{\alpha}{2} + \frac{\pi}{2}\right) = \left(\frac{1}{\sqrt{2}} \cos \frac{\alpha}{2} - \frac{1}{\sqrt{2}} \sin \frac{\alpha}{2}\right)^2$$

$$= \frac{1}{2} \left(\cos^2 \frac{\alpha}{2} + \sin^2 \frac{\alpha}{2} - 2 \cos \frac{\alpha}{2} \sin \frac{\alpha}{2} \right)$$

$$= \frac{1}{2} (1 - \sin \alpha) \quad \checkmark$$

Can also use $\varphi = \pi$, $\theta = \alpha$.

Q1) Zeeman Effect in H-atom

$$\hat{H} = \underbrace{\mu_B \vec{\sigma}_e \cdot \vec{\sigma}_p}_{\text{Spin-spin electron, proton}} - (\underbrace{\mu_B \vec{\sigma}_e \cdot \vec{B}}_{\text{Homogeneous } \vec{B}\text{-field.}} + \mu_p \vec{\sigma}_p \cdot \vec{B})$$

Start ✓ 3.1, 3.2

$$\vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2} \left[(\vec{S}_1 + \vec{S}_2)^2 - \vec{S}_1^2 - \vec{S}_2^2 \right]$$

$\nearrow S_1, S_2 = \frac{1}{2}$
 $\nwarrow \hbar^2 S(S+1) |_{S=0,1}$ (total spin)

$$S^2 |S, S_z\rangle = S(S+1) |S, S_z\rangle.$$

$$S_1 + S_2 = 0:$$

$$S_1^2 = \frac{1}{2} \left(\frac{1}{2} + 1 \right) \hbar^2 \Rightarrow$$

$$-S_1^2 - S_2^2 = -\hbar^2 \left(\frac{3}{4} + \frac{3}{4} \right)$$

$$\text{and } (\vec{S}_1 + \vec{S}_2) |S_1, S_{1z}; S_2, S_{2z}\rangle = 0.$$

and we obtain

$$\boxed{\vec{S}_1 \cdot \vec{S}_2 |S_1, S_{1z}; S_2, S_{2z}\rangle = -\hbar^2 \frac{3}{4} |S_1, S_{1z}; S_2, S_{2z}\rangle}$$

For $\vec{S}_1 + \vec{S}_2 = 2$, we have

$$(\vec{S}_1 \cdot \vec{S}_2) |S_1, S_{1z}; S_2, S_{2z}\rangle = \frac{\hbar^2}{2} \left(2 - \frac{6}{4} \right) = \frac{1}{4} \hbar^2.$$

3.2

$$S_{i\pm} = S_{ix} \pm iS_{iy}$$

$$\begin{aligned}\vec{S}_1 \cdot \vec{S}_2 &= S_{1z} S_{2z} + \frac{1}{2} \left([S_{1x} + iS_{1y}][S_{2x} - iS_{2y}] + [S_{1x} - iS_{1y}][S_{2x} + iS_{2y}] \right) \\ &= S_{1z} S_{2z} + \frac{1}{2} \left(S_{1x} S_{2x} - iS_{1x} S_{2y} + iS_{1y} S_{2x} + S_{1y} S_{2y} \right. \\ &\quad \left. + S_{1x} S_{2x} + S_{1y} S_{2y} + iS_{1x} S_{2y} - iS_{1y} S_{2x} \right) \\ &= S_{1z} S_{2z} + S_{1x} S_{2x} + S_{1y} S_{2y} \quad \checkmark\end{aligned}$$

1) Eigenvalues & eigenvectors of $\hat{H}$.

$$\begin{aligned}\hat{H} &= \beta \vec{\sigma}_c \cdot \vec{\sigma}_p - (\mu_c \vec{B} \cdot \vec{\sigma}_c + \mu_p \vec{B} \cdot \vec{\sigma}_p) \\ &\rightarrow \frac{\beta}{2} (\vec{\sigma}_c + \vec{\sigma}_p)^2 - \vec{\sigma}_c^2 - \vec{\sigma}_p^2 \\ \hat{S} &= \frac{\hbar}{2} \vec{\sigma}\end{aligned}$$

$$\vec{\sigma}_c \cdot \vec{\sigma}_p \big|_{s=0} = -\frac{3}{4} \hbar^2 \Rightarrow \vec{\sigma}_c \cdot \vec{\sigma}_p \big|_{s=0} = -3$$

$$\vec{\sigma}_c \cdot \vec{\sigma}_p \big|_{s=1} = \frac{\hbar^2}{4} \Rightarrow \vec{\sigma}_c \cdot \vec{\sigma}_p \big|_{s=1} = 1$$

Take $\vec{B} = B \hat{z}$ (no loss of generality)

$$\hat{H} = \beta \vec{\sigma}_c \cdot \vec{\sigma}_p - \mu_c B \sigma_3^{(c)} - \mu_p B \sigma_3^{(p)}$$

Total spin = 1 or 0 according to addition,

Use 4d Hilbert space

$$|11\rangle = |\frac{1}{2}, \frac{1}{2}\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle = |++\rangle$$

$$|10\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle)$$

$$|1-1\rangle = |--\rangle$$

$$|00\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle)$$

Compute Eigenvalues

$$H|11\rangle = [\beta \cdot 1 - \mu_e B - \mu_p B] |11\rangle$$

$$H|1-1\rangle = [\beta \cdot 1 + \mu_e B + \mu_p B] |1-1\rangle$$

$$\begin{aligned} H|10\rangle &= \beta \cdot 1 - \left(\mu_e B \sigma_{e,3} \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle) + \mu_p B \sigma_{p,3} \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle) \right) \\ &= \beta - \left(\mu_e B \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle) + \mu_p B \frac{1}{\sqrt{2}}(-|+-\rangle + |-+\rangle) \right) \end{aligned}$$

$$- \mu_e B \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle)$$

$|00\rangle$

$$= \beta - \mu_e B |00\rangle + \mu_p B |00\rangle.$$

$$H|00\rangle = -3\beta - \left(\mu_e B \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle) + \mu_p B \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle) \right)$$

$$= -3\beta - \left(\mu_e \beta \frac{1}{\sqrt{2}} (1+1+1+1) + \mu_p \beta \frac{1}{\sqrt{2}} (-1+2-1+2) \right) \\ - (\mu_e \beta |1,0\rangle - \mu_p \beta |1,0\rangle)$$

$$= -3\beta - \mu_e \beta |1,0\rangle + \mu_p \beta |1,0\rangle.$$

$$\begin{pmatrix} \langle 00 | H | 00 \rangle & \langle 00 | H | 10 \rangle \\ \langle 10 | H | 00 \rangle & \langle 10 | H | 10 \rangle \end{pmatrix} = \begin{pmatrix} -3\beta & \beta(\mu_p - \mu_e) \\ \beta(\mu_p - \mu_e) & \beta \end{pmatrix}$$

E's of this matrix

$$\det \begin{pmatrix} -3\beta - \lambda & \beta(\mu_p - \mu_e) \\ \beta(\mu_p - \mu_e) & \beta - \lambda \end{pmatrix} = -(3\beta + \lambda)(\beta - \lambda) - \beta^2(\mu_p - \mu_e)^2 \\ = -3\beta^2 + \lambda^2 + 3\beta\lambda - \beta\lambda - \beta^2(\mu_p - \mu_e)^2 \\ 0 = \lambda^2 + 2\beta\lambda - \beta^2(\mu_p - \mu_e)^2 - 3\beta^2.$$

$$\lambda = \frac{-2\beta \pm \sqrt{4\beta^2 - 4(1)(-\beta^2(\mu_p - \mu_e)^2 - 3\beta^2)}}{2}$$

$$= -\beta \pm \sqrt{\beta^2 + \beta^2(\mu_p - \mu_e)^2 + 3\beta^2}$$

$$= -\beta \pm \sqrt{4\beta^2 + \beta^2(\mu_p - \mu_e)^2}$$

"Hyperfine Splitting"

$$\begin{pmatrix} -2\beta \pm \sqrt{4\beta^2 + B^2(\mu_p - \mu_c)^2} & B(\mu_p - \mu_c) \\ B(\mu_p - \mu_c) & \mp \sqrt{4\beta^2 + B^2(\mu_p - \mu_c)^2} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Calculate $v_1, v_2 \dots$

Q3) $\frac{1}{2}_{(1)} \otimes \frac{1}{2}_{(2)} \otimes \frac{1}{2}_{(3)}$ system of spins
 $= \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}$

1) Construct $\frac{1}{2}_{(1)} \otimes \frac{1}{2}_{(2)} = 1 \oplus 0$.

$\bar{J}_1 = \frac{1}{2}$ $\bar{J}_2 = \frac{1}{2}$ $J_1 + J_2 = 1, 0$

$|11\rangle$

$|10\rangle$

$|00\rangle$

$|1-1\rangle$

$|11\rangle$ is max, both $\bar{J}_1, \bar{J}_2, m_1, m_2$ max

$$|11\rangle = \left| \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\begin{aligned} J_- |1, 1\rangle &= \hbar \sqrt{1(1+1) - 1(1-1)} |1, 0\rangle \\ &= \sqrt{2} \hbar |1, 0\rangle \end{aligned}$$

$$= (J_- | \frac{1}{2}, \frac{1}{2} \rangle) \otimes | \frac{1}{2}, \frac{1}{2} \rangle + | \frac{1}{2}, \frac{1}{2} \rangle \otimes (J_- | \frac{1}{2}, \frac{1}{2} \rangle)$$

$$= \hbar \sqrt{\frac{1}{2}(\frac{1}{2}-1) - \frac{1}{2}(\frac{1}{2}-1)} | \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, \frac{1}{2} \rangle + \hbar \sqrt{\frac{1}{2}(\frac{1}{2}-1) - \frac{1}{2}(\frac{1}{2}-1)} | \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{\sqrt{2}} \left| \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

Then, by orthogonality

$$|1,-1\rangle = \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

We calculate $|0,0\rangle$:

$$|0,0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{1}{\sqrt{2}} \left| \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

Now Construct $(1 \otimes 0) \otimes \frac{1}{2}_{(3)} = \overbrace{1 \otimes \frac{1}{2}_{(3)}}^{(i)} + \overbrace{0 \otimes \frac{1}{2}_{(3)}}^{(ii)}$
 problem 2

(i) is given by problem 2,

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle = |1,1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

All * given
below except
100

$$\left| \frac{3}{2}, \frac{1}{2} \right\rangle = \frac{1}{\sqrt{3}} |1,1\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} |1,0\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\left| \frac{3}{2}, -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} |1,0\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle + \frac{1}{\sqrt{3}} |1,-1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\left| \frac{3}{2}, -\frac{3}{2} \right\rangle = |1,-1\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\left| \frac{1}{2}, \frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} |1,1\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle - \frac{1}{\sqrt{3}} |1,0\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \frac{1}{\sqrt{3}} |1,0\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} |1,-1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$(ii) 0 \otimes \frac{1}{2} = \frac{1}{2}'$$

$$j_1=0 \quad j_2=\frac{1}{2} \quad \text{gives } m_1=\frac{1}{2} \text{ (max)} \\ m_1=-\frac{1}{2} \text{ (min)}$$

$$|\frac{1}{2}, \frac{1}{2}\rangle = |00; \frac{1}{2} \frac{1}{2}\rangle$$

$$|\frac{1}{2}, -\frac{1}{2}\rangle = |00; \frac{1}{2} -\frac{1}{2}\rangle$$

only 2 states, one
with max m
and one w/ min m

Can concatenate all of these together

$$|\frac{3}{2} \frac{3}{2}\rangle = |\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle$$

Any 2 spins interchangeable;
normalization remains

$$|\frac{3}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{3}} |\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\rangle + \frac{1}{\sqrt{3}} |-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle + \frac{1}{\sqrt{3}} |\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\rangle$$

$$|\frac{3}{2}, -\frac{1}{2}\rangle = \frac{1}{\sqrt{3}} |-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\rangle + \frac{1}{\sqrt{3}} |\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\rangle + \frac{1}{\sqrt{3}} |\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\rangle$$

$$|\frac{3}{2}, -\frac{3}{2}\rangle = |-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\rangle$$

Interchangeable $s_1 \leftrightarrow s_2$ spins
normalization remains

$$|\frac{1}{2}, \frac{1}{2}\rangle = \sqrt{\frac{2}{3}} |\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\rangle - \frac{1}{\sqrt{6}} |-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle - \frac{1}{\sqrt{6}} |\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\rangle$$

$$|\frac{1}{2}, -\frac{1}{2}\rangle = \frac{1}{\sqrt{6}} |-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\rangle + \frac{1}{\sqrt{6}} |\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\rangle - \sqrt{\frac{2}{3}} |-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\rangle$$

$$|\frac{1}{2}, \frac{1}{2}\rangle' = \frac{1}{\sqrt{2}} |\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\rangle + \frac{1}{\sqrt{2}} |-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle$$

$$|\frac{1}{2}, -\frac{1}{2}\rangle' = \frac{1}{\sqrt{2}} |\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\rangle + \frac{1}{\sqrt{2}} |-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle \quad \text{degeneracy?}$$

b) Show that $|\frac{1}{2}, m_{\frac{1}{2}}\rangle$ and $|\frac{1}{2}, m_{\frac{1}{2}}\rangle'$ states are eigenstates of $\vec{S}_1 \cdot \vec{S}_2$ operator. Show that $\vec{S}$ operators can be used to distinguish between the two.

Notice that $\vec{S}_1 \cdot \vec{S}_2$ acting on $|\frac{1}{2}, m_{\frac{1}{2}}\rangle'$ gives

$$0 \otimes \frac{1}{2}' = -\frac{3}{4}\hbar^2$$

Since $0 = 0_{i+2}$, $\vec{S}_i$ is 1, 2

$\vec{S}_1 \cdot \vec{S}_2$ on $|\frac{1}{2}, m_{\frac{1}{2}}\rangle$ gives $\frac{1}{4}$, and

on $|\frac{3}{2}, m_{\frac{3}{2}}\rangle$ gives $\frac{1}{4}$, so can

distinguish between the two.

This is because $(\vec{S}_1 + \vec{S}_2 + \vec{S}_3)^2 = \vec{S}_{\text{tot}}^2$

$$= \underbrace{\vec{S}_1^2 + \vec{S}_2^2 + \vec{S}_3^2}_{\substack{3/4 \text{ each}}} + 2\vec{S}_1 \cdot \vec{S}_2 + 2\vec{S}_1 \cdot \vec{S}_3 + 2\vec{S}_2 \cdot \vec{S}_3$$

$\rightarrow 2 \cdot \begin{cases} -3/4 \text{ on } \frac{1}{2}' \\ \frac{1}{4} \text{ on } \frac{1}{2}, \frac{3}{2} \end{cases}$

So $\vec{S}_i \cdot \vec{S}_j$ commutes with $\vec{S}_{\text{tot}}^2$

can use both to classify states.

